Online Form: RC, RL and RLC Circuits - Differential Equations Using Experimental Activities

Dear students:

This questionnaire is part of a study to assess the use of experimental activities in theoretical courses at the State Distance University (UNED) of Costa Rica.

To obtain relevant and essential information for the research objectives regarding students' perspectives on using technological tools such as Remote Laboratories (RLs), which provide a real experience through remote access over the Internet, we kindly ask you to answer the questions below.

Please also note that the information you provide will be treated confidentially and will serve as a basis for this study.

Thank you in advance for your valuable cooperation.

Instructions

The questionnaire consists of three parts, as follows:

I. General information.

II. Use of experimental activities.

III. Final summary.

I. General Information

The following questions seek to gather general information about participants who completed Experimental Activities during the first semester of 2024 at UNED, using the HIVE laboratory to illustrate the use of Differential Equations in Engineering.

1. Age:
2. Before completing the experimental activity, had you used experimental tools to better understand a topic in theoretical courses? (yes-or-no question)
3. If you answered Yes to the previous question, use the space below to describe your previous experience with experimental activities:
4. Which university center do you belong to?
5. Which device(s) (Laptop, Tablet, Cell Phone, Desktop Computer, other) did you use to access the Remote Laboratory?

II. Use of experimental activities.

The following three blocks ask you to answer the question: What advantages does the resource used in the experimental work offer?

In each block, indicate your level of agreement or disagreement with each statement, using the following scale: 1. Strongly disagree; 2. Disagree; 3. Neither disagree nor agree; 4. Agree; 5. Strongly agree.

The blocks are divided to gather your views on Usability, Learning Perception and Satisfaction during the different Experimental Activities.

6. Usability.

Statement	1	2	3	4	5
I found the experimental activities on Differential Equations applied to RC, RL and RLC circuits, using the HIVE Remote Laboratory, easy to complete.	○	○	○	○	○
During the experimental activities on RC, RL and RLC circuits, I felt confident using the HIVE Remote Laboratory and was able to follow all the instructions without any problems.	○	○	○	○	○
The information provided in the manual for carrying out the experimental activities on RC, RL and RLC circuits helped me access and adjust the conditions within the HIVE Remote Laboratory.	○	○	○	○	○
The information provided in the manual for presenting the solutions to the experimental activities on RC, RL and RLC circuits is easy to understand, was useful to me and is complete.	○	○	○	○	○
I would like more experimental activities that illustrate the course content to be used in other mathematics-related courses.	○	○	○	○	○
Being able to access the HIVE Remote Laboratory at any time and as often as I wanted, in order to follow the instructions for the experimental activities on RC, RL and RLC circuits, helped me complete these activities successfully.	○	○	○	○	○
This type of experimental resource meets my needs as a distance education student.	○	○	○	○	○

7. Learning Perception

Statement	1	2	3	4	5
The experimental activities on RC, RL and RLC circuits, using the HIVE Remote Laboratory, helped me better understand the process of solving Differential Equations.	○	○	○	○	○
The experimental activities on RC, RL and RLC circuits helped me better understand the applications of Differential Equations in technology.	○	○	○	○	○
This type of activity introduced me to experimental work through presenting, manipulating and organizing experimental data.	○	○	○	○	○
The experimental activities I completed helped me improve my skills in designing and presenting representations of experimental data and interpreting those data using tables and graphs.	○	○	○	○	○
The experimental activities improved my ability to visualize and relate the application of Differential Equations to examples involving technology, giving me a better understanding of the content covered in class.	○	○	○	○	○
During the experimental activities on RC, RL and RLC circuits, I was able to follow all the instructions in the manual, including collecting data, performing calculations and answering questions.	○	○	○	○	○

8. Satisfaction.

Statement	1	2	3	4	5
Overall, I am satisfied with the experimental activities I completed.	○	○	○	○	○
The experimental activities were relevant to my learning in this course.	○	○	○	○	○
I would like other experimental activities to be offered in this course.	○	○	○	○	○
I felt motivated by the experimental activities in this course.	○	○	○	○	○

9. How do you assess this type of educational resource? (open-ended question)

10. What would you recommend to improve the instructions and solutions manual? (open-ended question)

III. Final summary.

Please answer the following question by selecting a point on the scale according to your opinion. The position of each of your responses indicates how closely your choice aligns with the adjectives at either end.

What is your opinion of the experimental activity carried out in the differential equations course using a deferred laboratory?

11.

0	1	2	3	4	5	6	7	8	9	10

0: Unpleasant 10: Pleasant

12.

0	1	2	3	4	5	6	7	8	9	10

0: Boring 10: Fun

13.

0	1	2	3	4	5	6	7	8	9	10

0: Indifferent 10: Interesting

14.

0	1	2	3	4	5	6	7	8	9	10

0: Difficult 10: Easy

15.

0	1	2	3	4	5	6	7	8	9	10

0: The time spent is not appropriate 10: The time spent is appropriate

16.

0	1	2	3	4	5	6	7	8	9	10

0: The instructions are unclear 10: The instructions are clear

17.

0	1	2	3	4	5	6	7	8	9	10

0: Does not improve understanding of Differential Equations.

10: Improves understanding of Differential Equations.